Standard Errors and Confidence Intervals for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

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Abstract

Mediation modeling using longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in longitudinal data, they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the direct, indirect, and total effects, for a range of time-interval values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the direct, indirect, and total effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and Monte Carlo methods to calculate standard errors and confidence intervals for the direct, indirect, and total effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

Keywords: mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

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Mediation is a widely adopted model across the social, behavioral, and health sciences. A recent Google Scholar search, conducted as of 24 January 2024, using search terms `"indirect effect OR mediation AND social science OR behavioral science OR health science"`, yielded over five million entries. While many mediation models still rely on cross-sectional data, where time is ignored (Figure 1a), there is a growing trend toward employing longitudinal designs, where time is implicitly considered in the model (e.g., Figures 1b and 1c) addressing the critical issue of temporal ordering, thus, enhancing the plausibility of causal interpretations regarding mediation effects (Cole & Maxwell, 2003; Georgeson et al., 2023; Maxwell & Cole, 2007; Maxwell et al., 2011; O'Laughlin et al., 2018). Mediation models help us uncover the intermediate steps through which an independent variable X (such as an intervention) influences a dependent variable Y (such as an outcome) through one or more mediator variables M (e.g., Baron & Kenny, 1986; MacKinnon, 2008). By identifying these mediating variables, we gain insights into the direct and indirect causal pathways that connect interventions to their effects (Fairchild & MacKinnon, 2014; MacKinnon, 1994; Mackinnon & Dwyer, 1993; O'Rourke & MacKinnon, 2018, 2019).

The aim of this manuscript is to provide methods to quantify and visualize the uncertainty associated with point estimates of direct, indirect, and total effects as they change across a range of time-interval values using the continuous-time vector autoregressive (CT-VAR) framework. We do this with repeated measurements over time as a stronger basis for causal inference[1] in contrast to cross-sectional designs (Gollob & Reichardt, 1987, 1991). Furthermore, we take the continuous-time framework as previous

[1] For observational data, we limit our treatment of causality following the definition provided by Granger (1969), that is, when one variable, denoted as X, Granger-causes another variable, denoted as Y, it implies that past values of X contain information that helps predict future values of Y beyond what can be predicted using only past values of Y itself.

research (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) have shown its benefits particularly in addressing the time-interval dependency problem, that is, direct, indirect, and total effect sizes change as a function of the time-interval used to measure the variables. Our novel contributions to this area of research include introducing the delta and Monte Carlo methods as ways to quantify the uncertainty associated with point estimates of direct, indirect, and total effects across a range of time-intervals. Furthermore, we use the delta and Monte Carlo methods to generate regions of significance which help in visualizing for which range of time-interval values the direct, indirect, and total effects are statistically different from zero. Lastly, the methods we developed as well as visualization tools are made available in a free and open-source `R` package `cTMed`.

This manuscript is divided into the following sections: (1) introduction of continuous-time mediation and the need to quantify uncertainty associated with mediation effects; (2) hypothetical examples where we introduce point estimates for continuous-time mediation effects, delta and Monte Carlo method confidence intervals around point estimates, and regions of significance across a range of time-intervals; (3) simulation study examining the finite sample performance of the delta and Monte Carlo method confidence intervals; and (4) application of the delta and Monte Carlo method confidence intervals on an empirical data set. We close with a summary of our novel contributions, limitations of the current work, and future directions.

Mediation Using Longitudinal Data

In this section we discuss the rationale for moving from discrete-time to continuous-time mediation. We provide an overview of the continuous-time modeling framework, namely the linear stochastic differential equation model. While application of the continuous-time framework for mediation models exist in the literature (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021), quantifying the uncertainty associated with mediation effects are yet to be developed. The last part of this section outlines how we address this research gap.

From Discrete-Time to Continuous-Time Mediation

Mediation and causality have always been linked. From the "causal steps" approach (e.g., Baron & Kenny, 1986; James & Brett, 1984; Judd & Kenny, 1981) to the longitudinal framework (e.g., Cole & Maxwell, 2003; Gollob & Reichardt, 1987, 1991; Maxwell & Cole, 2007; Maxwell et al., 2011), the idea is that causality operates not only directly ($X \rightarrow Y$), but also indirectly ($X \rightarrow M \rightarrow Y$). Gollob and Reichardt (1987, 1991) proposed three principles of causality namely (1) "causes take time to exert their effects and, therefore, causal variables must occur before outcome variables"; (2) "variables can have effects in themselves"; (3) "size of the effect typically varies with the length of time lag".

Principles 1 and 2 underscore the fact that cross-sectional mediation presented in Figure 1a cannot capture the direct and indirect causal associations between X, M, and Y and the need to move to a model where previous values of variables are used to predict their current values as in the autoregressive cross-lagged mediation model presented in Figures 1b and 1c. The term autoregressive (AR) refers to allowing the variables to have effects in themselves and the term cross-lagged or crossregressive (CR) refers to using previous values of other variables to predict current values of a variable. The umbrella term used for these types of models is discrete-time vector autoregressive model (DT-VAR). The term vector autoregressive (VAR) means that there are more than one variable in the system. The term discrete-time (DT) means that we observe the variables at equally spaced time intervals (de Haan-Rietdijk et al., 2017). DT-VAR encompasses panel models where the observations are made in a few time points as well as data from intensive longitudinal data collection where many repeated observations are made.

Principle 3 underscores the time-interval dependency, that is, in DT-VAR, effect sizes change as a function of the time-interval used to measure the variables (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021; Wang & Zhang, 2020). This means that, given the same underlying causal process, the moment in time we used to observe the variables of interest (e.g., daily, monthly, yearly) influences the size of the effects we observe.

In a cross-sectional mediation, if X changes by one unit, then Y changes by τ' units, holding M constant. This is the direct effect of X on Y. If X changes by one unit, then M changes by α units as a result. Similarly, if M changes by one unit, then Y changes by β units as a result. The resulting change of α units on M produced by a unit change in X would then produce a corresponding change of β units on Y. Thus, Y changes by $\alpha\beta$ units through M as X changes by one unit. This is the indirect effect. The total effect is the sum of direct and indirect effects. In DT-VAR, this idea of decomposing effects also holds. However, we are tracing effects from previous values of the variables to the next. The indirect effect of X at time t on Y at time $t+2$ is the product of the path from X at time t to M at time $t+1$ and the path from M at time $t+1$ to Y at time $t+2$. While many mediation models still rely on cross-sectional data, the good news is that there is a growing trend toward employing DT-VAR models[2]. However, the indirect effects and all the other effects we observe in DT-VAR changes on account of the time-interval we use to observed the variables. A change in modeling framework from discrete-time to continuous-time can address this issue.

The challenges associated with DT-VAR can be addressed by adopting a continuous-time vector autoregressive (CT-VAR) modeling framework. CT-VAR parameters are defined at any time-interval, as such we can obtain the direct, indirect, and total effects for any time-interval. From this framework, we can explore how effects change as a function of time-interval. Additionally, unlike DT-VAR, which assumes equally spaced data points, CT-VAR accommodates unequally spaced data–a common occurrence in intensive longitudinal studies. These models allow us to capture the evolving dynamics of parameters across a range of time intervals (Boker, 2002; Oud & Delsing, 2010; van Montfort et al., 2018; Voelkle et al., 2012). However, embracing a continuous time perspective necessitates a conceptual shift. Specifically, lagged regression parameters, that

[2] See the decrease in cross-sectional studies from the systematic review by Maxwell and Cole (2007) to O'Laughlin et al. (2018).

is the AR and CR parameters, for a given time-interval within this framework are interpreted as total effects, rather than direct effects (Aalen et al., 2012, 2016; Deboeck & Preacher, 2015; Ryan & Hamaker, 2021).

The Linear Stochastic Differential Equation (SDE) Framework

In our modeling approach, we consider processes that continuously evolve over time. This means that the variables can assume values at any given moment. To capture this behavior, we employ linear stochastic differential equations (SDEs). Specifically, we focus on a special type of SDE known as the Ornstein–Uhlenbeck model (Chow et al., 2023; Oravecz et al., 2011; Uhlenbeck & Ornstein, 1930).

Similar to structural equation modeling, we can break down our model into two key components: the measurement component and the structural component. The measurement model describes the relationship between observed variables and latent (unobserved) variables. Specifically, consider the following equation:

$$\begin{aligned} \mathbf{y}_{i,t_m} &= \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_{i,t_m} + \boldsymbol{\varepsilon}_{i,t_m}, \\ \text{with} \quad \boldsymbol{\varepsilon}_{i,t_m} &\sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Theta}\right) \end{aligned} \tag{1}$$

where $\mathbf{y}_{i,t_m}$ represents a vector of observed random variables, $\boldsymbol{\eta}_{i,t_m}$ a vector of latent random variables, and $\boldsymbol{\varepsilon}_{i,t_m}$ a vector of normally distributed random measurement errors, for time t at measurement occasion m and individual i. $\boldsymbol{\nu}$ denotes a vector of intercepts, $\boldsymbol{\Lambda}$ a matrix of factor loadings, and $\boldsymbol{\Theta}$ the covariance matrix of $\boldsymbol{\varepsilon}$. In the context of a mediation model with three variables where the effect the variable X on the variable Y is mediated by the variable M, we define $\mathbf{y}$ and $\boldsymbol{\eta}$ as follows:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix}, \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix}. \tag{2}$$

The dynamic structure is given by

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = -\boldsymbol{\Phi}\left(\boldsymbol{\mu} - \boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t} \tag{3}$$

where $\boldsymbol{\mu}$ is the equilibrium or mean value, $\boldsymbol{\Phi}$ is the drift matrix, $\boldsymbol{\Sigma}$ is the matrix of volatility or randomness in the process (also known as the diffusion matrix), and $\mathrm{d}\boldsymbol{W}$ is a Wiener process or Brownian motion, which represents random fluctuations. The Ornstein–Uhlenbeck process centers around a central tendency denoted by $\boldsymbol{\mu}$. When the process deviates from this mean, it naturally gravitates back toward it. We can think of $\boldsymbol{\mu}$ as an "attractor" that pulls the process back to its typical value. When we specify the model such that the latent state variables have a stable mean vector $\boldsymbol{\mu}$ consisting of zeros, Equation 3 simplifies to

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = \left(\boldsymbol{\Phi}\boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t}. \tag{4}$$

The diagonal elements of the matrix $\boldsymbol{\Phi}$ (referred to as auto effects) control the speed at which the process reverts to its own equilibrium value, $\boldsymbol{\mu}$. When the absolute value of these elements is high, the variable quickly returns to its mean after being perturbed. Smaller absolute values result in slower reversion, allowing the variable to exhibit more persistent behavior away from its mean. The off-diagonal elements of $\boldsymbol{\Phi}$ represent associations between variables and their rates of change. A negative cross effect indicates that if one variable takes on a positive value, there is a resulting decrease in another variable's value. The magnitude of this effect is determined by the absolute value. The diagonal elements of the matrix $\boldsymbol{\Sigma}$ determine the magnitude of volatility around the central tendency $\boldsymbol{\mu}$. Larger values indicate greater fluctuations or noise in the process, while smaller values result in smoother trajectories. Essentially, $\boldsymbol{\Sigma}$ quantifies the degree of randomness or shock-induced variability. The off-diagonal elements of $\boldsymbol{\Sigma}$ represent correlations between random shocks across variables in the system. In summary, the Ornstein–Uhlenbeck process is a mean-reverting stochastic process. Over time, it drifts

toward its equilibrium value $\boldsymbol{\mu}$, experiences volatility (as captured by $\boldsymbol{\Sigma}$), and reverts back to the mean with a rate determined by $\boldsymbol{\Phi}$. For the context of this manuscript, we refer to the Ornstein–Uhlenbeck model as the first-order continuous-time vector autoregressive model (CT-VAR).

The drift matrix $\boldsymbol{\Phi}$ holds particular significance in the context of mediation models. It serves as a conduit for understanding the associations between the states of the variables under investigation. Notably, within a trivariate model, involving η_X, η_M, and η_Y, numerous paths from one variable to another can be explored. This comprehensive connectivity arises because we can construct the model such that all variables are interconnected. Consequently, beyond the classic mediation model $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$, we can also investigate alternative mediation pathways, such as $\eta_Y \rightarrow \eta_M \rightarrow \eta_X$, using the same data set. For our current focus, we will delve into the specific $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ path. In the subsequent section, we will explore the relationship between CT-VAR and DT-VAR models, emphasizing the pivotal role played by the drift matrix $\boldsymbol{\Phi}$ within the $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ mediation model.

Link Between CT-VAR and DT-VAR

To understand the connection between CT-VAR and DT-VAR models, we need to examine the discrete-time state space model parameterization of the CT-VAR model which is the discrete-time solution in integral form given by

$$\begin{aligned} \boldsymbol{\eta}_{i,t_{m_i}} &= \boldsymbol{\alpha}_{\Delta t_{m_i}} + \boldsymbol{\beta}_{\Delta t_{m_i}} \boldsymbol{\eta}_{i,t_{m_i-1}} + \boldsymbol{\zeta}_{i,t_{m_i}}, \\ \text{with} \quad \boldsymbol{\zeta}_{i,t_{m_i}} &\sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{m_i}}\right) \end{aligned} \tag{5}$$

where $\boldsymbol{\eta}_{i,t_{m_i}}$ and $\boldsymbol{\eta}_{i,t_{m_i-1}}$ represent the vector of latent variables in the current and previous measurement occasion m while $\boldsymbol{\zeta}_{i,t_{m_i}}$ is the normally distributed error term in the current measurement occasion. The model parameter $\boldsymbol{\alpha}$ denotes a vector of intercepts, $\boldsymbol{\beta}$ a matrix of coefficients describing the impact of the lagged values $\boldsymbol{\eta}_{i,t_{m-1}}$ on the current value $\boldsymbol{\eta}_{i,t_m}$ (AR coefficients on the diagonal and CR coefficients on the off-diagonal), and $\boldsymbol{\Psi}$ the

covariance matrix of $\boldsymbol{\zeta}$. Notice that the parameters have subscript Δt, representing that their values are dependent on the time-interval, that is, given the same continuous process, their values change depending on the time-interval.

In the context of CT-VAR models, the underlying process can take values at any arbitrary time point. However, our observations occur discretely. This means that while the process itself operates continuously, we only observe it at specific discrete time instances. Let's compare this to DT-VAR, where observations are made at specific discrete time intervals (usually positive integers). In a DT-VAR example, suppose we collect data every hour from 7 AM to 10 AM for four measurement occasions. We make discrete observations in time t, scaled by the hour, on specific measurement occasions $m = \{1, \cdots, 4\}$. These occasions correspond to time points: $t_1 = 7$, $t_2 = 8$, $t_3 = 9$, and $t_4 = 10$. The time-interval from one measurement occasion to the next is constant and equal to 1 hour. In the CT-VAR model, observations can occur at any time between 7 AM and 10 AM for the same four measurement occasions. Specifically, the time points are given by: $t_1 = 7.166667$ (equivalent to 7:10 AM), $t_2 = 8.283333$ (equivalent to 8:17 AM), $t_3 = 9.45$ (equivalent to 9:27 AM), and $t_4 = 10$ (equivalent to 10:00 AM). Notably, the time-interval from one measurement occasion to the next varies allows for irregularly spaced observations. We denote this time-interval as Δt, representing the difference between the current measurement occasion m and the previous measurement occasion $m - 1$. From here we can see that if Δt is constant as in DT-VAR, the model is simply a CT-VAR model where Δt is constant. As such, the DT-VAR model is a special case of the CT-VAR model.

Let us look at the link between the CT-VAR and the discrete-time solution parameters. Whereas, the parameter $\boldsymbol{\Phi}$ describes the moment-to-moment association of one variable on the rate of change in another, the parameter $\boldsymbol{\beta}$ is a matrix of lagged parameters, that is, the effect of the variable on a previous measurement occasion to the current measurement occasion, given a particular time-interval Δt. The discrete-time $\boldsymbol{\beta}$ can be obtained from the continuous-time $\boldsymbol{\Phi}$ using

$$\boldsymbol{\beta} = \exp\left(\Delta t \boldsymbol{\Phi}\right). \tag{6}$$

Furthermore, the discrete-time parameter $\boldsymbol{\alpha}$ is related to the continuous-time parameters $\boldsymbol{\Phi}$ and $\boldsymbol{\mu}$

$$\begin{aligned} \boldsymbol{\alpha} &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\exp\left(\Delta t \boldsymbol{\Phi}\right) - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right) \\ &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\boldsymbol{\beta} - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right). \end{aligned} \tag{7}$$

If we solve for $\boldsymbol{\mu}$ in Equation 7, we arrive at

$$\boldsymbol{\mu} = \left(\boldsymbol{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}. \tag{8}$$

Assuming weak stationarity, the mean vector of $\boldsymbol{\eta}$ as a function of the discrete-time parameters $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ is given by $\boldsymbol{\mu}_{\boldsymbol{\eta}} = \left(\mathbf{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}$. This is equivalent to the CT-VAR parameter $\boldsymbol{\mu}$ as a function of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ given by Equation 8. Lastly, we can convert the continuous-time process noise covariance matrix $\boldsymbol{\Sigma}$ to the discrete-time counter part $\boldsymbol{\Psi}$[3]. Note that $\mathbf{J}$ and $\mathbf{I}$ are identity matrices and $\text{vec}\left(\cdot\right)$ stacks the values from a matrix into a vector by column.

$$\begin{aligned} \text{vec}\left(\boldsymbol{\Psi}\right) &= \mathbf{C}\left[\exp\left(\Delta t \mathbf{C}\right) - \mathbf{J}\right]\text{vec}\left(\boldsymbol{\Sigma}\right) \\ \text{with} \quad \mathbf{C} &= \left(\boldsymbol{\Phi} \otimes \mathbf{I}\right) + \left(\mathbf{I} \otimes \boldsymbol{\Phi}\right) \end{aligned} \tag{9}$$

In summary, we can see the DT-VAR parameters are related to CT-VAR parameters. We can also see from the discrete state space integral solution of CT-VAR

[3] The notation presented here is mathematically equivalent to the notation presented in the works of Arnold (1974), Oud and Jansen (2000), and Voelkle and Oud (2012). We prefer this notation because it facilitates straightforward calculations in statistical packages that support standard linear algebra operations.

that DT-VAR is a special case of CT-VAR where the time-interval Δt is constant across time and individuals. As Δt is part of the calculation of the parameters α, β, and Ψ, their values change as a function of the time-interval.

Quantifying Uncertainty in Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

The work of Deboeck and Preacher (2015) provided a starting point for structuring mediation processes in CT-VAR. However, their work is far from complete: complete inferences require the availability of reasonable standard error estimates and ways of quantifying the uncertainty around the direct, indirect, and total effects in mediation in continuous time. In the mediation literature, it is not enough to have point estimates of the indirect effects. It is important to quantify the uncertainty associated with it. Although the idea of the indirect effect as a product of regression coefficients is straightforward, quantifying the uncertainty of the point estimates has some complications and much of the mediation literature has focused on this task. The sampling distribution of the product of normally distributed random variables is not necessarily normal (Aroian, 1947; Craig, 1936), leading to complexities in computing p-values for the indirect effect. Because of this, confidence intervals have been used to quantify uncertainty for the point estimates. Using the delta method, confidence intervals that are asymptotically normally distributed have been proposed (Aroian, 1947; Goodman, 1960; Sobel, 1982, 1986, 1987). Alternative approaches that take into account the asymmetry have also been proposed including: (1) The distribution of the product method, which analytically examines the sampling distribution of the product of normally distributed coefficients (MacKinnon et al., 2004, 2007); (2) Nonparametric bootstrapping, involving resampling the original data and fitting the model multiple times to create a sampling distribution of the indirect effect (Bollen & Stine, 1990; Preacher & Hayes, 2008; Shrout & Bolger, 2002); (3) Parametric bootstrapping, involving simulating raw data using the estimated parameters and fitting the model similar to nonparametric bootstrapping (Preacher & Selig, 2012); (4) The Monte

Carlo method, which simulates regression coefficients to generate a sampling distribution of the indirect effect (MacKinnon et al., 2004; Pesigan & Cheung, 2023; Preacher & Selig, 2012); (5) Profile likelihood-based approach that estimate confidence intervals using the likelihood ratio test (Cheung, 2009a, 2009b; Pawitan, 2013; Pesigan & Cheung, 2020; Venzon & Moolgavkar, 1988). From these approaches, nonparametric bootstrap has become the preferred method (Bollen & Stine, 1990; Preacher & Hayes, 2008; Shrout & Bolger, 2002). However, the multiple model fitting especially for more complex models can be a bottleneck making it a less viable approach (see Pesigan & Cheung, 2023, for the case of mediation models with missing data). The Monte Carlo method has been shown to be as good as nonparametric bootstrap in a number of situations (Pesigan & Cheung, 2023; Tofighi & Kelley, 2019, 2020). Because the empirical sampling distributions of parameter estimates are generated and not obtained via multiple model fitting, the Monte Carlo method is computationally more efficient than nonparametric bootstrapping.

In this manuscript, drawing from the mediation literature (e.g., MacKinnon et al., 2004; Pesigan & Cheung, 2023; Preacher & Selig, 2012; Sobel, 1986), we provide estimation procedures to complete the inferential process in CT-VAR mediation using the delta and Monte Carlo methods. Additionally, we provide the `R` package `cTMed` that calculates point estimates, standard errors, and confidence intervals for the direct, indirect, and total effects. The package also provides plotting tools to visualize how the effects change given a range of time-intervals.

Hypothetical Examples

In this section, we use hypothetical examples within the CT-VAR framework to (1) elaborate on what we mean by direct, indirect, and total effects within this modeling framework, (2) develop the delta and Monte Carlo methods for generating standard errors and confidence intervals for the effects previously defined, and (3) introduce regions of significance using the delta and Monte Carlo methods to visualize statistically significant results across a range of time-intervals.

Hypothetical Example 1: Understanding Direct, Indirect, and Total Mediation Effects in CT-VAR Models

In both cross-sectional and autoregressive models, the fundamental concept of mediation analysis involves decomposing the total effect into direct and indirect components. This approach allows us to understand how the mediator variable M influences the relationship between X and Y. However, in the context of autoregressive models, several authors (e.g., Aalen et al., 2016; Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) propose a conceptual shift when thinking about the lagged parameters $\boldsymbol{\beta}$. In this conceptualization, $\boldsymbol{\beta}$ is treated as the matrix of total effects rather than direct effects. This shift has implications for thinking about mediation models in VAR models particularly when it comes to decomposing the total effect into direct and indirect effects. In the following hypothetical example, we use simulated data from descriptive statistics reported by Grundy et al. (2007) to demonstrate how the direct and indirect effects are decomposed from the total effect. We used the `mvrnorm` function from the `MASS R` package (Venables & Ripley, 2002) to generate multivariate normal data using the correlation matrix reported in Table 1 of Grundy et al. (2007). The researchers tested the hypothesis that maternal knowledge mediated the association between mother reported marital conflict and child reported behavioral competence. Furthermore, they added a reciprocal effect from behavioral competence to knowledge. The original data set was collected using a discrete time-interval of a year over a period of four years. Note that Deboeck and Preacher (2015) also use the descriptive statistics from Grundy et al. (2007) to demonstrate continuous-time mediation. While our approach is similar to that of Deboeck and Preacher (2015), for illustrative purposes, the model we fitted has a number of differences. First, we fitted a CT-VAR model using the simulated data set where both the measurement and the dynamic structures are estimated. Second, we specified all possible paths from one variable to another to be estimated in the drift matrix $\boldsymbol{\Phi}$. Third, we increased the sample size from 133 to 500 to increase the statistical power to detect

significant effects. Prior to fitting the model in the `dynr` `R` package (Ou et al., 2019), we inserted missing values in between measurement occasions discrete measurement occasions using .10 increments to aid in the optimization process.

We label the independent variable marital conflict X, the mediator variable maternal knowledge M, and dependent variable behavioral competence Y. Since, we are incorporating a measurement model, the latent variables η_X, η_M, and η_Y represent the latent variables in the model with X, M, and Y as observed indicators. The estimated CT-VAR drift matrix $\mathbf{\Phi}$ is presented in Table 1. In the study, time t is scaled in years and $\Delta t = 1$ a time-interval of one year between measurement occasions given by Figure 2a. Given Equation 6 we can calculate the matrix of lagged coefficients for a time-interval $\Delta t = 1$ from the drift matrix $\mathbf{\Phi}$ as follows:

$$\begin{aligned} \boldsymbol{\beta}_{\Delta t(1)} &= \exp\left(\mathbf{\Phi}\right) \\ &= \begin{array}{c} \\ \eta_{X_{t+1}} \\ \eta_{M_{t+1}} \\ \eta_{Y_{t+1}} \end{array} \begin{array}{c} \begin{array}{ccc} \eta_{X_t} & \eta_{M_t} & \eta_{X_t} \end{array} \\ \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \beta_{1,3} \\ \beta_{2,1} & \beta_{2,2} & \beta_{2,3} \\ \beta_{3,1} & \beta_{3,2} & \beta_{3,3} \end{pmatrix} \end{array}. \end{aligned} \tag{10}$$

Now we look at the case of a DT-VAR model from the same continuous process except that the time-interval used is three years ($\Delta t = 3$) between measurement occasions. From the time-series literature (e.g., Hamilton, 1994) we can derive $\boldsymbol{\beta}_{\Delta t(3)}$ from $\boldsymbol{\beta}_{\Delta t(1)}$ using matrix powers. The key is to raise the matrix with the smaller time-interval by the factor in which the time-interval increased to the larger time-interval. For example, the change from $\Delta t = 1$ to $\Delta t = 3$ is a factor of three, thus $\boldsymbol{\beta}^3_{\Delta t(1)} = \boldsymbol{\beta}_{\Delta t(3)}$. Therefore, we can say that $\boldsymbol{\beta}_{\Delta t(3)} = \exp\left(3\mathbf{\Phi}\right) = \boldsymbol{\beta}^3_{\Delta t(1)}$. Let us explore the relationship of the elements of $\boldsymbol{\beta}_{\Delta t(3)}$ as a function of the elements of $\boldsymbol{\beta}_{\Delta t(1)}$. In particular, we will focus on the third row and first column in $\boldsymbol{\beta}_{\Delta t(3)}$ pertaining to the association of marital conflict (η_X) at time t

and behavioral competence (η_Y) at time $t+3$. The purple line in Panel 2a in Figure 2 represents the total effect of marital conflict (η_X) at time t on behavioral competence (η_Y) at time $t+3$. This total effect is decomposed as a function of the lagged parameters in $\boldsymbol{\beta}_{\Delta t(1)}$. By raising Equation 10 to the third power and subsetting the third row and the first column of the resulting matrix we get

$$\begin{aligned}\text{Total} &= \boldsymbol{\beta}_{\Delta t(3)}\,[3,1] \\ &= \beta_{1,1}\beta_{1,1}\beta_{3,1} + \beta_{2,1}\beta_{1,2}\beta_{3,1} + \beta_{3,1}\beta_{1,3}\beta_{3,1} \\ &+ \beta_{1,1}\beta_{2,1}\beta_{3,2} + \beta_{2,1}\beta_{2,2}\beta_{3,2} + \beta_{3,1}\beta_{2,3}\beta_{3,2} \\ &+ \beta_{1,1}\beta_{3,1}\beta_{3,3} + \beta_{2,1}\beta_{3,2}\beta_{3,3} + \beta_{3,1}\beta_{3,3}\beta_{3,3},\end{aligned} \tag{11}$$

where each term represents distinct paths from marital conflict (η_X) at time t to behavioral competence (η_Y) at time $t+3$ graphically represented by Panels 2b to 2j in Figure 2. Note that red paths in are paths where maternal knowledge (η_M) is traversed. We define these red paths as indirect effects of marital conflict (η_X) at time t on behavioral competence (η_Y) at time $t+3$ through maternal knowledge (η_M). The sum of these red paths constitute the overall indirect effect. The blue paths are the direct effects and their sum constitutes the overall direct effect of marital conflict (η_X) at time t on behavioral competence (η_Y) at time $t+3$. The sum of the direct and indirect effects represents the total effect of marital conflict (η_X) at time t on behavioral competence (η_Y) at time $t+3$.

Now we divide any arbitrary time-interval Δt into infinitely many smaller parts and insert these infinitely many latent variables between values at time t and time $t+\Delta t$. The idea in the CT-VAR approach is that the lagged parameters from the discrete-time integral solution represents the sum of all the effects from the infinitely many smaller parts in between time t and time $t+\Delta t$. The decomposition we discussed earlier still apply. How do we then obtain the direct, indirect, and total effects? The total effect is straight forward as it is simply the result of exponentiating the product of the drift matrix and the

time-interval. Using the estimated drift matrix presented in Table 1 and a time-interval $\Delta t = 3$, the total effect matrix is given by

$$\begin{aligned}
\text{Total} &= \exp\left(\Delta t \boldsymbol{\Phi}\right) \\
&= \exp\left[3\begin{pmatrix} -0.034 & 0.038 & -0.013 \\ -0.152 & -0.586 & 0.228 \\ -0.100 & 0.104 & -0.512 \end{pmatrix}\right] \\
&= \begin{pmatrix} 0.892 & 0.049 & -0.006 \\ -0.240 & 0.183 & 0.139 \\ -0.172 & 0.056 & 0.238 \end{pmatrix}.
\end{aligned} \tag{12}$$

Following Ryan and Hamaker (2021), the matrix of direct effects is obtained by removing the effects in the drift matrix $\boldsymbol{\Phi}$ that originates from and/or goes to the mediator (or mediators) before exponentiating. We do this by pre- and post-multiplying the drift matrix $\boldsymbol{\Phi}$ with a diagonal matrix $\mathbf{D_m}$ where the ith and j^{th} index pertaining to the mediators are set to zero. In our current example where maternal knowledge (η_M) is treated as the mediator variable, the direct effect matrix is given by

$$\begin{aligned}
\text{Direct} &= \exp\left(\Delta t \boldsymbol{D_m} \boldsymbol{\Phi} \boldsymbol{D_m}\right) \\
&= \exp\left[3\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -0.034 & 0.038 & -0.013 \\ -0.152 & -0.586 & 0.228 \\ -0.100 & 0.104 & -0.512 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right] \\
&= \begin{pmatrix} 0.908 & 0.000 & -0.019 \\ 0.000 & 1.000 & 0.000 \\ -0.145 & 0.000 & 0.218 \end{pmatrix}.
\end{aligned} \tag{13}$$

The indirect effect can now be obtained by subtracting the direct effect matrix from the total effect matrix.

$$
\begin{aligned}
\text{Indirect} &= \text{Total} - \text{Direct} \\
&= \exp\left(\Delta t \boldsymbol{\Phi}\right) - \exp\left(\Delta t \boldsymbol{D_m} \boldsymbol{\Phi} \boldsymbol{D_m}\right) \\
&= \begin{pmatrix} -0.016 & 0.049 & 0.013 \\ -0.240 & -0.817 & 0.139 \\ -0.027 & 0.056 & 0.021 \end{pmatrix}
\end{aligned} \tag{14}
$$

We subset the ith row and j^{th} column of the resulting matrices for the direct, indirect, and total effects of the mediator variable/s on the path from the jth column variable (1 for marital conflict in our case) to the ith row variable (3 for behavioral competence in our case). The direct, indirect, and total effects of marital conflict (η_X) on behavioral competence (η_Y) through maternal knowledge (η_M), for a time-interval of 3, are -0.145, -0.027, and -0.172, respectively.

The `Med` function in the `cTMed R` package calculates the direct, indirect, and total effects for a specific time-interval or a range of time-intervals. A plot method for the `Med` function can be used to visualize the effects as a function of time-interval. See https://jeksterslab.github.io/cTMed/reference/Med.html for more details.

Hypothetical Example 2: Quantifying Uncertainty in Direct, Indirect, and Total Mediation Effects in CT-VAR Models

In the previous section, we showed how to calculate point estimates for the direct, indirect, and total effect for a given time-interval from the CT-VAR drift matrix $\boldsymbol{\Phi}$. In particular, we know the direct, indirect, and total effects of marital conflict (η_X) on behavioral competence (η_Y) through maternal knowledge (η_M), for a time-interval of 3, are -0.145, -0.027, and -0.172, respectively. However, we still do not have a way to assess

these effects from an inferential point of view. In this section, we extend the previous section by providing methods for quantifying the uncertainty in the direct, indirect, and total effects we have defined. In the following sections, we introduce the delta and the Monte Carlo methods. We chose these approaches because they are relatively easy to compute even for complex models. We also wish to contrast the performance of a method where symmetry is imposed versus and a method where asymmetry is accounted for. We discuss these methods in general and then apply them to the context of generating standard errors and confidence intervals as a way to quantify the uncertainty for the direct, indirect, and total effects.

The Delta Method

The delta method is a powerful statistical technique used to derive the asymptotic distribution of a random variable. It becomes particularly handy when dealing with functions of other random variables that are asymptotically Gaussian, that is, their distributions converge to a Gaussian distribution as sample size increases. As previously mentioned, it has been applied in psychological research to generate standard errors and confidence intervals for the indirect effect (Aroian, 1947; Goodman, 1960; Sobel, 1986, 1987). It has also been applied to other functions of parameters such as the standardized regression coefficients (Jones & Waller, 2015; Pesigan et al., 2023; Yuan & Chan, 2011). Currently, popular SEM software packages such as `Mplus` (Muthén & Muthén, 2017) and `lavaan` (Rosseel, 2012) use the delta method as the standard approach to generating standard errors and confidence intervals for defined functions of SEM parameters. In `R`, the `Delta` and `DeltaGeneric` functions in the `betaDelta` (Pesigan et al., 2023) package can generate standard errors and confidence intervals for functions of parameters of fitted models as long as the estimates and the sampling variance-covariance matrix are provided.

We define the delta method following the multivariate central limit theorem. Let $\boldsymbol{\theta}$ be $\operatorname{vec}(\boldsymbol{\Phi})$, that is, the elements of the drift matrix $\boldsymbol{\Phi}$ in vector form sorted column-wise. Let $\hat{\boldsymbol{\theta}}$ be $\operatorname{vec}\left(\hat{\boldsymbol{\Phi}}\right)$, that is the estimates of the elements of the drift matrix. The function

$\mathbf{g}$using $\hat{\boldsymbol{\theta}}$ as input can be expressed as:

$$\sqrt{n}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)-\mathbf{g}\left(\boldsymbol{\theta}\right)\right)\xrightarrow{\mathrm{D}}\mathcal{N}\left(0,\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right) \tag{15}$$

where $\mathbf{J}$ is the matrix of first-order derivatives of the function $\mathbf{g}$ with respect to the elements of $\boldsymbol{\theta}$ and $\boldsymbol{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\boldsymbol{\theta}}$. From the former, we can derive the distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ for a sample size n as follows

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)\approx\mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right),n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right). \tag{16}$$

The uncertainty associated with the estimator $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is, therefore, given by $n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'$. When $\boldsymbol{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\boldsymbol{\theta}}$, that is, $\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)$ for $n^{-1}\boldsymbol{\Gamma}$. Therefore, the sampling variance-covariance matrix of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is given by

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)\approx\mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right),\mathbf{J}\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\mathbf{J}'\right). \tag{17}$$

From this general approach, we can generate the sampling variance-covariance matrix of any function of the drift matrix $\boldsymbol{\Phi}$. The standard errors, that is, the square-root of the diagonal elements of the sampling variance-covariance matrix is used to generate symmetric confidence intervals around the point estimates of any function of the drift matrix $\boldsymbol{\Phi}$. In current application, let the function $\mathbf{g}$ be a vector of the direct, indirect, and total effects of a specific independent variable on a dependent variable through one or more mediator variables. The `DeltaMed` function in the `cTMed` package applies the delta method to generate standard errors, test statistics, p-values and confidence intervals for these effects for a specific time-interval or a range of time-intervals. See https://jeksterslab.github.io/cTMed/reference/DeltaMed.html for more details.

The Monte Carlo Method

The Monte Carlo method was introduced in the mediation literature by MacKinnon et al. (2004) and formalized by Preacher and Selig (2012) as a resampling technique for generating confidence intervals for the indirect effect. It has been shown to be as good as nonparametric bootstrap (Pesigan & Cheung, 2023; Tofighi & Kelley, 2019, 2020; Tofighi & MacKinnon, 2015). As a resampling technique, it is similar to the bootstrap such that the sampling distribution of the function of parameters does not need to be normal. Unlike the bootstrap, however, it is more computationally efficient as it does not involve multiple model fitting. The `R` package `semmcci` provides a general approach to applying the Monte Carlo method for generating confidence intervals for a function of any SEM parameters. The `MCFunc` and `MCGeneric` functions in the `semmcci` (Pesigan & Cheung, 2023) package can generate standard errors and confidence intervals for functions of parameters of fitted models as long as the estimates and the sampling variance-covariance matrix are provided.

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters.

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}\left(\boldsymbol{\theta}, \mathbb{V}\left(\hat{\boldsymbol{\theta}}\right)\right) \tag{18}$$

Using this distributional assumption, a sampling distribution of $\hat{\boldsymbol{\theta}}$ which we refer to as $\hat{\boldsymbol{\theta}}^*$ can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\boldsymbol{\theta}}^* \sim \mathcal{N}\left(\hat{\boldsymbol{\theta}}, \hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\right). \tag{19}$$

A sampling distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$, which we refer to as $\mathbf{g}\left(\hat{\boldsymbol{\theta}}^*\right)$, can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the standard errors. In resampling approaches, it is customary to use the percentiles of the generated distribution as confidence intervals instead of the standard errors. This approach allows the confidence limits to be asymmetric around the point estimates unlike

the delta method which is always symmetric. Percentiles corresponding to $100\,(1-\alpha)\,\%$ are the confidence intervals.

From this general approach, we can generate the empirical sampling distribution of any function of the drift matrix $\mathbf{\Phi}$. There are constraints, however, on the randomly generated values of $\mathbf{\Phi}$. First, the diagonal values should be from zero to negative infinity. Second, the real part of all eigenvalues is less than zero. These two can be implemented using a replacement sampling approach, that is, discarding randomly generated values where the two constraints do not hold. Again, let the function $\mathbf{g}$ be a vector of the direct, indirect, and total effects of a specific independent variable on a dependent variable through one or more mediator variables. The `MCMed` function in the `cTMed` package applies the Monte Carlo method to generate the standard errors and confidence intervals for these effects for a specific time-interval or a range of time-intervals. See https://jeksterslab.github.io/cTMed/reference/MCMed.html for more details.

Confidence Intervals for the Direct, Indirect and Total Effects

Using the delta and Monte Carlo methods as implemented by the `cTMed` package, the standard errors and confidence intervals for the direct, indirect, and total effects of marital conflict (η_X) on behavioral competence (η_Y) through maternal knowledge (η_M), for time-intervals Δt of 1, 2, and 3 are presented in Table 2. We chose these time-intervals as they are the time-intervals available in the data set. Based on the results we highlight a few observations. The direct, indirect, and total effects changes across time-intervals. In fact, the absolute value of the direct, indirect, and total effects increase as time-intervals increase implying stronger effects at larger time gaps between measurements. Looking at the confidence intervals, however, we can see that the direct and total effects are not statistically different from zero while the indirect effect is significant. We can see this as the confidence intervals for the indirect effect for all three time-intervals did not contain zero. From a substantive point of view, this shows that the association between marital conflict (η_X) and behavioral competence (η_Y) is mediated by maternal knowledge (η_M)

given time-intervals of one, two, and three years.

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Table 1

Point and Interval Estimates of the CT-VAR Drift Matrix $\mathbf{\Phi}$ *for Data Simulated from Grundy et al. (2007)*

Parameter	Estimate	Standard Error	2.5%	97.5%
Conflict to conflict ($X \to X$)	-0.034	0.012	-0.056	-0.011
Conflict to knowledge ($X \to M$)	-0.152	0.031	-0.213	-0.090
Conflict to competence ($X \to Y$)	-0.100	0.029	-0.157	-0.043
Knowledge to conflict ($M \to X$)	0.038	0.019	0.000	0.075
Knowledge to knowledge ($M \to M$)	-0.586	0.091	-0.764	-0.408
Knowledge to competence ($M \to Y$)	0.104	0.055	-0.004	0.212
Competence to conflict ($Y \to X$)	-0.013	0.017	-0.047	0.020
Competence to knowledge ($Y \to M$)	0.228	0.059	0.113	0.343
Competence to competence ($Y \to Y$)	-0.512	0.075	-0.659	-0.364

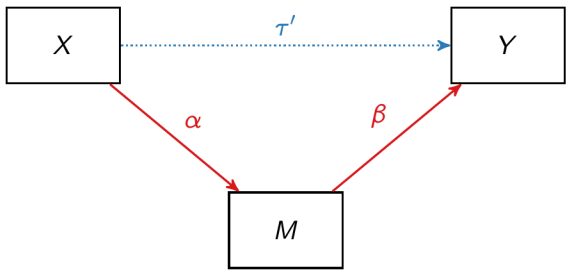
Table 2

Point, Standard Errors, and Interval Estimates of the Direct, Indirect, and Total Effects for Data Simulated from Grundy et al. (2007)

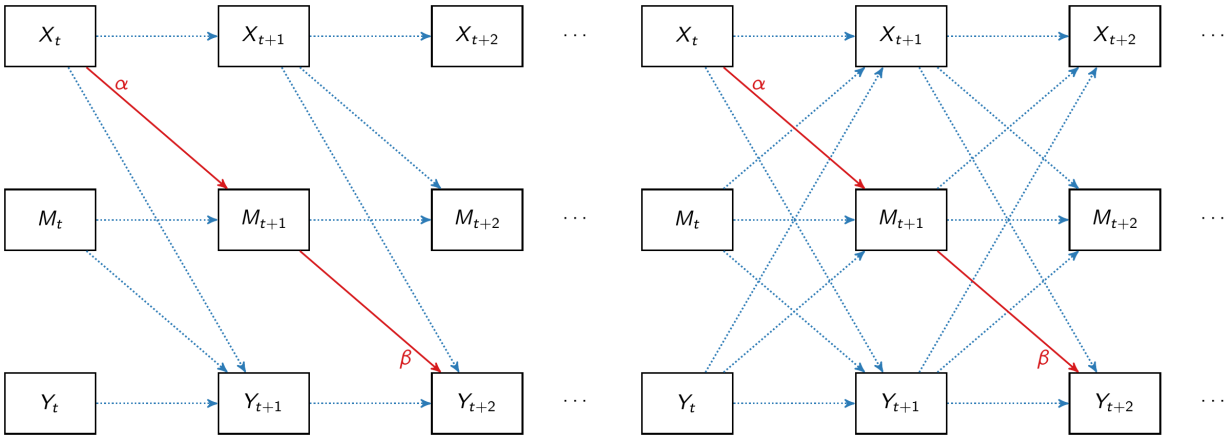
Effect	Estimate	Delta Method			Monte Carlo Method		
		Standard Error	2.5%	97.5%	Standard Error	2.5%	97.5%
			$\Delta t = 1$				
Direct	-0.077	0.023	-0.121	-0.033	0.023	-0.122	-0.033
Indirect	-0.006	0.003	-0.012	0.000	0.003	-0.012	0.000
Total	-0.083	0.022	-0.126	-0.040	0.022	-0.126	-0.039
			$\Delta t = 2$				
Direct	-0.121	0.036	-0.191	-0.051	0.036	-0.193	-0.051
Indirect	-0.016	0.009	-0.034	0.001	0.009	-0.034	0.001
Total	-0.137	0.035	-0.205	-0.070	0.035	-0.205	-0.069
			$\Delta t = 3$				
Direct	-0.145	0.044	-0.231	-0.059	0.044	-0.235	-0.061
Indirect	-0.027	0.014	-0.055	0.001	0.015	-0.055	0.003
Total	-0.172	0.042	-0.254	-0.090	0.042	-0.254	-0.090

Figure 1

Cross-Sectional and Longitudinal Mediation Models



(a) *Cross-sectional mediation model.*



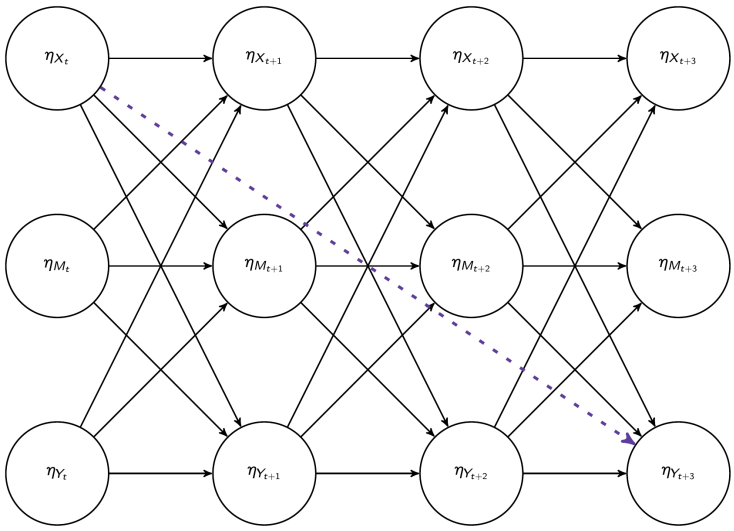
(b) *Unidirectional autoregressive cross-lagged mediation model.*

(c) *Bidirectional autoregressive cross-lagged mediation model.*

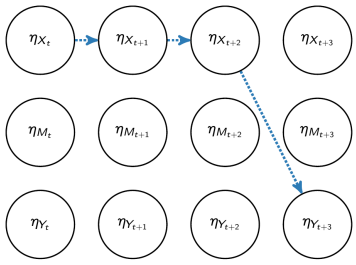
Note: The indirect effect of M on the path from X to Y is the product of α and β. t, $t+1$, and $t+2$ represent equally spaced time points. The dots indicate that the model can accommodate more time points. In the bidirectional autoregressive cross-lagged mediation model, the path from the dependent variable to the independent and the mediator variables as well as the path from the mediator variable to the independent are included in the model. These paths are omitted in the unidirectional model.

Figure 2

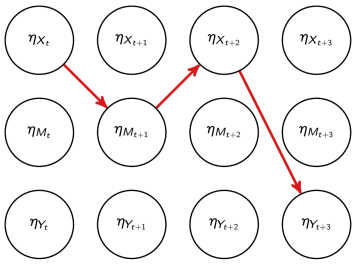
Decomposing the Direct and Indirect Effects from the Total Effect (Bidirectional)



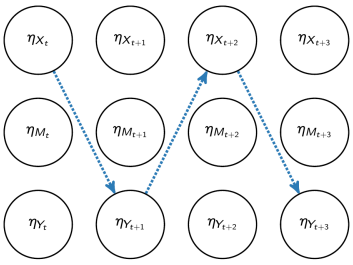
(a) *DT-VAR with time-interval of one. The purple line represents the total effect of X on Y at a time-interval of three.*



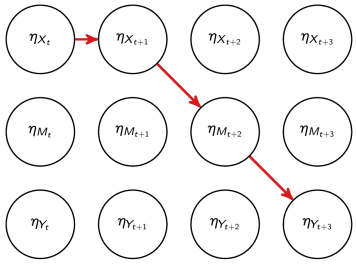
(b) $\beta_{1,1}\beta_{1,1}\beta_{3,1}$



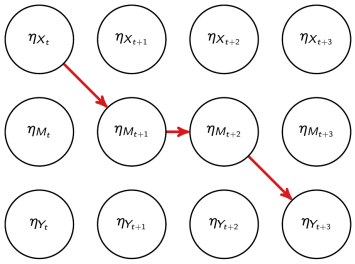
(c) $\beta_{2,1}\beta_{1,2}\beta_{3,1}$



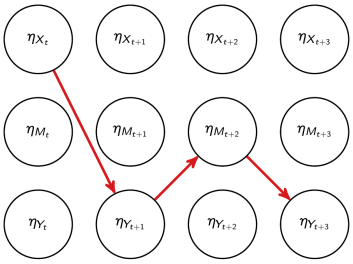
(d) $\beta_{3,1}\beta_{1,3}\beta_{3,1}$



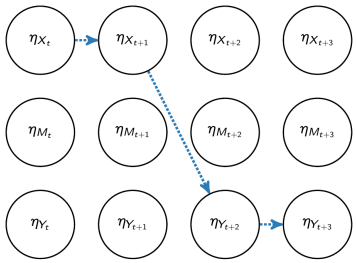
(e) $\beta_{1,1}\beta_{2,1}\beta_{3,2}$



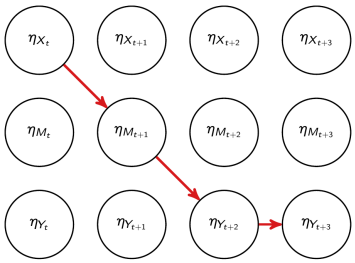
(f) $\beta_{2,1}\beta_{2,2}\beta_{3,2}$



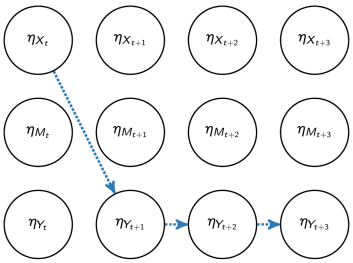
(g) $\beta_{3,1}\beta_{2,3}\beta_{3,2}$



(h) $\beta_{1,1}\beta_{3,1}\beta_{3,3}$



(i) $\beta_{2,1}\beta_{3,2}\beta_{3,3}$



(j) $\beta_{3,1}\beta_{3,3}\beta_{3,3}$

Note: Panel (2a) represents the total effect of η_X on η_Y at a time-interval of three is a function of the lagged parameters for a time-interval of one. Panels (2b) to (2j) represent the decomposition of the total effect (purple) into direct (blue) and indirect (red) effects.